

Report
Of
Applying Community Detection to Find Habitat Clusters

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: 25SCS1003004017

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Batch: 2026-27

Internship: Online Summer Internship in Data Science and Machine Learning

UpSkill Campus / UniConverge Technologies

25 July 2026 - 25 August 2026

CANDIDATE'S DECLARATION

I, **SACHIN KUMAR**, hereby declare that the internship report titled “**Applying Community Detection to Find Habitat Clusters**” has been prepared by me to fulfil the academic requirement of the Bachelor of Technology in Computer Science and Engineering. The report documents the project work and learning undertaken during my Data Science and Machine Learning internship.

The project presents a graph-based approach to habitat analysis. Habitat locations are represented as nodes, relationships between locations are represented as weighted edges, and community detection is applied to discover groups with stronger internal connectivity.

Where an illustrative or synthetic dataset is used, results are treated as demonstrations of the analytical workflow rather than as field-validated ecological conclusions.

Student Name	SACHIN KUMAR
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Program	B.Tech CSE
Project Topic	Applying Community Detection to Find Habitat Clusters
Internship Domain	Data Science and Machine Learning

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported me during my internship in Data Science and Machine Learning. The internship provided a structured setting in which I could study data preparation, exploratory analysis, graph-based modeling, machine learning concepts, and practical project implementation.

I am thankful to **Unstop** for the internship opportunity and learning environment. I also acknowledge the instructional and mentoring support associated with the internship, which helped me connect classroom concepts with a complete project workflow.

I am grateful to the **School of Computer Science and Engineering, IILM University, Greater Noida**, for providing the academic framework within which this internship report has been prepared.

Finally, I thank my teachers, mentors, friends, and family members for their encouragement and support throughout the internship. The project strengthened my understanding of how graphs, algorithms, and visualization can be combined to study relationships in complex datasets.

Student Name	Sachin Kumar
Roll No.	25SCS1003004017

INTERNSHIP COMPLETION CERTIFICATE

The internship completion certificate supplied for Sachin Kumar is reproduced below as supporting evidence. It records successful completion of the internship from **25-07-2026 to 25-08-2026** in the domain of **Data Science and Machine Learning**, with collaboration involving AICTE, UniConverge Technologies and upSkill Campus.



Certificate ID: USC640790TIA · Certificate issued to Sachin Kumar

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PROJECT DESCRIPTION

Project Title: Applying Community Detection to Find Habitat Clusters

Internship Domain: Data Science and Machine Learning

Internship Mode: Online

Internship Period: 25 July 2026 to 25 August 2026

Training Platform / Organization: Unstop

The project demonstrates how graph analytics can be used to identify habitat communities from structured habitat observations. Instead of treating every habitat record as an isolated row, the approach creates a network in which relationships between locations carry analytical meaning.

4.1 Introduction

Community detection is a graph-analysis task in which nodes are grouped according to the strength and pattern of their connections. For habitat analysis, this can help reveal areas that are more strongly related through spatial proximity, ecological similarity, shared environmental conditions, or observed species overlap.

The workflow follows dataset inspection, cleaning, feature preparation, similarity calculation, graph construction, community detection, cluster evaluation, visualization, and interpretation. The emphasis is on understanding the method rather than claiming that a single clustering result represents ecological truth.

4.2 Organization Profile

For this report, **Unstop** is identified as the internship organization/platform associated with the student's internship presentation. The internship context is Data Science and Machine Learning, with the project emphasizing practical analytical implementation.

The supplied completion certificate separately records the internship domain and the collaborating entities shown on the certificate. Those details are preserved in the certificate section rather than being rewritten on the issued certificate.

4.3 Problem Statement

Habitat observations are often stored as independent records containing geographic, environmental, and ecological attributes. Looking only at individual rows can hide relationships between locations. A more informative representation is to construct a graph in which each habitat is a node and an edge represents a meaningful relationship between two habitats.

The problem is therefore: **given habitat observations and measurable relationships between them, can community-detection algorithms identify groups of habitats that are more strongly connected to one another than to the rest of the network?**

The desired output is a set of habitat clusters that can be visualized and examined using graph-level measures such as modularity and community size.

4.4 Project Objectives

1. Prepare and inspect a structured habitat dataset.
2. Represent habitat locations as nodes in a graph.
3. Define meaningful weighted relationships between habitat locations.
4. Apply a community-detection algorithm to identify habitat clusters.
5. Visualize the resulting network and communities.
6. Evaluate cluster structure using graph-based measures.
7. Interpret the clusters cautiously and document limitations.
8. Create a reproducible Python workflow suitable for further ecological analysis.

4.5 Scope of the Project

The scope includes data preprocessing, habitat similarity modeling, graph construction, community detection, cluster visualization, and evaluation. The project is designed as an academic demonstration of graph analytics.

The project does not attempt to perform live wildlife tracking, species population forecasting, satellite-image segmentation, or production-grade conservation decision making. Those tasks would require additional datasets, domain validation, field observations, and operational safeguards.

4.6 Technologies and Tools Used

Tool / Technology	Purpose
Python	Core programming language for data analysis and graph processing
Pandas	Data loading, cleaning, aggregation and tabular analysis
NumPy	Numerical operations and feature calculations
NetworkX	Graph construction, connectivity and community analysis
Matplotlib	Network and analytical visualizations
Scikit-learn	Feature scaling and similarity-related preprocessing
Jupyter Notebook	Interactive experimentation and documentation
Git / GitHub	Version control and reproducibility
VS Code	Development and project organization

Table 4.1: Project technology stack

4.7 System Architecture

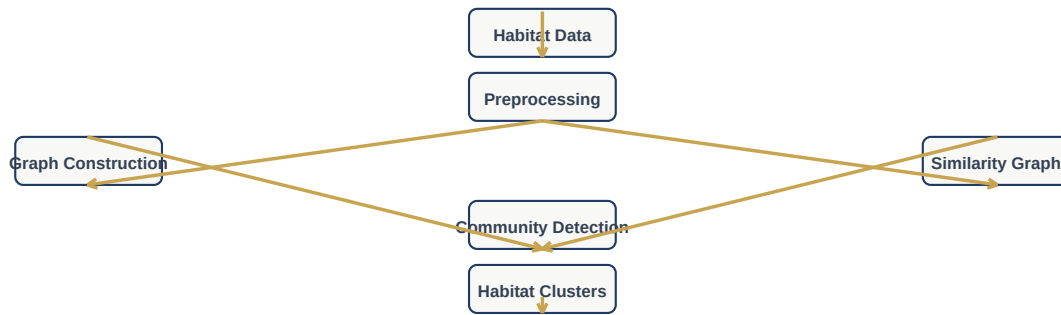


Figure 4.1: End-to-end system architecture

The system begins with structured habitat observations. After validation and preprocessing, habitat similarity is calculated and used to construct a weighted graph. Community detection then partitions the graph into groups. The final stage produces cluster summaries and visualizations for interpretation.

A graph representation is useful because it preserves relationships that are difficult to express through a simple table. For example, two habitats may be geographically close and environmentally similar even when their raw feature values are not identical.

4.8 Methodology

Step 1 — Dataset preparation: load habitat observations, check data types, handle missing values, remove duplicate records, and standardize relevant numerical attributes.

Step 2 — Relationship definition: calculate a similarity or distance measure between habitat records. A threshold can be used to avoid weak, noisy edges.

Step 3 — Graph construction: create one node per habitat and a weighted edge for sufficiently strong relationships.

Step 4 — Community detection: apply an algorithm such as Louvain to identify communities according to graph structure.

Step 5 — Evaluation: inspect modularity, community sizes, connectivity patterns, and visual structure.

4.8.1 Dataset Preparation

For a reproducible academic implementation, the project can use a structured habitat dataset containing location, environmental, and ecological attributes. If a field-validated dataset is unavailable, a synthetic dataset can be used to test the complete graph workflow. Synthetic data should be clearly labeled as such and should not be treated as ecological evidence.

A representative schema for the project is shown below.

Attribute	Description
habitat_id	Unique habitat identifier
latitude / longitude	Geographic location of the habitat
elevation	Approximate elevation or terrain measure
temperature	Representative environmental temperature measure
rainfall	Representative precipitation measure
vegetation_index	Numerical indicator of vegetation condition
water_distance	Approximate distance to the nearest water source
species_richness	Observed species richness or an index
forest_cover	Percentage or index of surrounding forest cover
region	Broad geographic or administrative grouping

Table 4.2: Habitat dataset attributes

The exact choice of variables depends on the available habitat dataset. The key requirement is that selected features have a defensible relationship to habitat similarity or connectivity.

4.8.2 Exploratory Data Analysis

Exploratory analysis is used to understand distributions, identify missing or extreme values, and determine which attributes may contribute to meaningful habitat relationships. Geographic plots can also reveal whether observations are concentrated in particular regions.

4.8.2 Exploratory Data Analysis — Visual Summary

Illustrative habitat observations by region

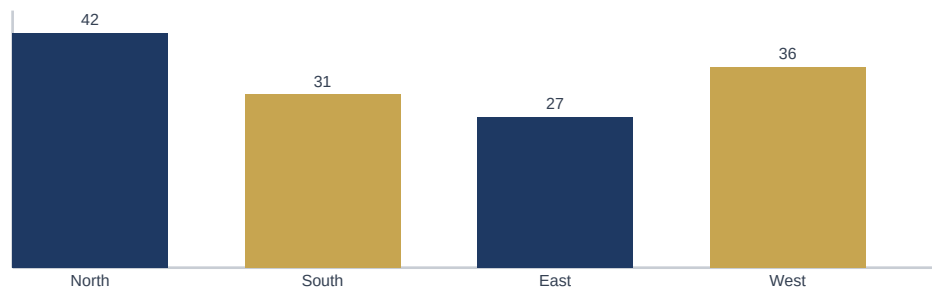


Figure 4.3: Habitat observations by region (illustrative example)

Illustrative distribution of pairwise similarity scores

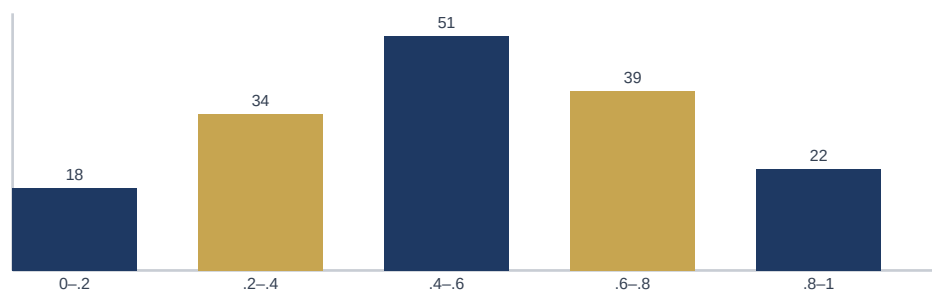


Figure 4.4: Similarity/connection distribution (illustrative example)

The charts demonstrate the type of exploratory analysis used in the workflow. They are not presented as measurements from a specific ecological field study unless the underlying dataset is supplied and validated.

A useful EDA stage also checks whether a small number of variables dominate the similarity calculation. Standardization can prevent attributes with larger numeric scales from overwhelming the graph-building process.

4.8.3 Preprocessing

Preprocessing is important because community detection operates on the graph produced by the similarity definition. Poorly scaled variables or missing values can create misleading edges and therefore misleading communities.

Numerical features can be standardized using a z-score transformation. Categorical attributes such as region can be encoded where appropriate, although inclusion should be justified because it can introduce artificial boundaries into the graph.

Duplicate habitat records should be removed or consolidated. Missing values should be handled according to the meaning of each feature rather than filled blindly. Geographic coordinates should also be checked for impossible or inconsistent values.

4.8.4 Graph Construction

Let each habitat be represented by a node v_i . For two habitats i and j , a similarity score $s(i,j)$ can be calculated from selected standardized features. An edge is added when the similarity exceeds a chosen threshold τ . The edge weight w_{ij} is the similarity score.

This produces a weighted graph $G = (V, E, W)$, where V is the set of habitats, E is the set of relationships, and W contains relationship strengths.

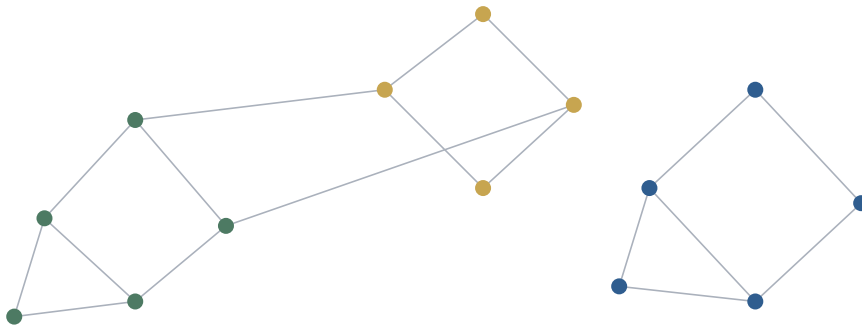
Threshold selection is a critical design choice. A very low threshold can create an overly dense graph; a very high threshold can fragment the graph and produce many tiny communities.

4.8.5 Community Detection

Community detection attempts to divide the graph into groups whose internal connectivity is relatively strong compared with connectivity across groups. For this project, the Louvain method is a suitable candidate because it is designed for modularity-based community detection on weighted networks.

The method iteratively moves nodes between communities when the move improves the modularity objective, then aggregates the resulting groups and repeats the process. The output is a community label for each habitat node.

These labels can be used to summarize cluster size, geographic distribution, average environmental characteristics, and internal connectivity.



Conceptual habitat network with three detected communities.

Node = habitat location | Edge = similarity/connectivity | Node group = community

Figure 4.2: Habitat network and detected communities (conceptual illustration)

A detected community should not automatically be called an ecological “habitat type.” Community detection finds network structure; ecological meaning must be established through domain interpretation and, ideally, independent validation.

4.8.6 Cluster Interpretation

If geographic coordinates dominate the relationship definition, detected communities may largely reflect spatial proximity. If environmental variables dominate, communities may instead represent environmental similarity. This distinction should be stated explicitly when interpreting the result.

4.8.7 Community Evaluation

Community-detection evaluation measures

Modularity	Within-community density	Higher often means clearer separation
Community count	Number of groups	Describes segmentation level
Conductance	Boundary connectivity	Lower can indicate stronger separation
Stability	Repeatability across settings	Stable communities are more credible

Table 4.3: Community-detection evaluation measures

Modularity is useful for assessing whether detected communities have stronger internal connectivity than expected under a reference structure. It is not, however, a universal measure of ecological validity.

Additional checks can include community-size balance, geographic coherence, stability under different similarity thresholds, and sensitivity to feature selection. Repeating the analysis under reasonable parameter changes helps identify whether a cluster is stable or merely an artifact of one configuration.

4.8.8 Community Size Comparison

Illustrative number of habitats per detected community

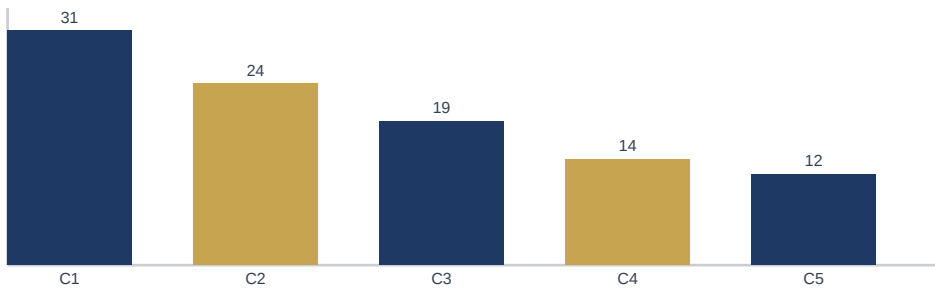


Figure 4.5: Community size comparison (illustrative example)

A strongly imbalanced community-size distribution is not automatically wrong, but it should be examined. Very small communities may represent isolated habitats, noisy data, or an overly strict graph threshold.

4.9 Expected Outcomes and Results

The primary expected outcome is a reproducible workflow that converts habitat observations into a network and identifies candidate communities. The result should include a community label for each habitat, a network visualization, community-size summaries, and evaluation measures.

Outcome	Result / Evidence
Working graph pipeline	Data preparation, similarity calculation and graph construction workflow
Community detection	Community labels generated using a graph community-detection method
Cluster visualization	Network visualization highlights community membership
Evaluation	Modularity and community-size analysis included
Reproducibility	Parameters and preprocessing steps documented
Interpretation	Cluster patterns described with explicit limitations

Table 4.4: Key project outcomes

Because the supplied source report does not contain a verified habitat dataset or measured community-detection output, this revised report deliberately avoids inventing a specific modularity score, cluster count, or ecological finding. Such values should be inserted after the actual project is executed.

4.9.1 Interpretation of Results

The useful result of community detection is not simply the number of clusters. The analysis should explain why detected communities are connected, which features contribute to their similarity, and whether the groups remain stable when reasonable parameters are changed.

The project demonstrates the connection between data preparation, graph construction, algorithmic community detection, visualization, and interpretation. That end-to-end workflow is the main learning outcome.

4.9.2 Limitations

The largest limitation is dataset validity. A synthetic or purely illustrative dataset cannot establish real ecological patterns. Even with a real dataset, sampling bias, spatial autocorrelation, missing observations, measurement error, and feature-selection choices can influence the graph.

Community-detection algorithms also have parameter and resolution effects. Different algorithms or thresholds may produce different partitions. Therefore, cluster membership should be treated as an analytical result that requires validation, not as an unquestionable ecological label.

4.9.3 Future Enhancements

Future work can replace illustrative data with a verified ecological dataset; incorporate GIS layers and protected-area boundaries; compare Louvain with Leiden and other community-detection methods; perform parameter sensitivity analysis; add spatial statistics; and validate clusters against expert ecological knowledge or independent observations.

A stronger implementation could provide an interactive map showing habitat locations, detected communities, environmental summaries, and cluster-level statistics.

4.10 CERTIFICATES OF COMPLETION AND SUPPORTING PROOF

The internship completion certificate is included in Section 3 of this report in proper full-page format. The supplied certificate records Sachin Kumar's successful completion of the internship from **25-07-2026 to 25-08-2026** in the domain of **Data Science and Machine Learning**.

The certificate also records collaboration involving **AICTE**, **UniConverge Technologies**, and **upSkill Campus**. The issued certificate itself has been reproduced without altering its content.

5. BIBLIOGRAPHY / REFERENCES

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7. Python Software Foundation. Python 3 Documentation.

Final Project Topic: Applying Community Detection to Find Habitat Clusters

Student: Sachin Kumar · **Roll No.:** 25SCS1003004017

INTERNSHIP



CERTIFICATE OF COMPLETION

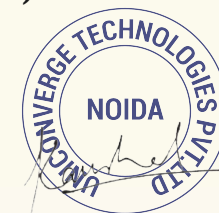
awarded to

Sachin Kumar

for successfully completing the Internship Program from **25-07-2026** to **25-08-2026**.

During this internship, the candidate actively participated in hands-on training and real industry project work in the domain of **Data Science And Machine Learning**

This program was conducted in collaboration with **All India Council for Technical Education (AICTE)** and **UniConverge Technologies & upSkill Campus**.



Certificate ID: USC640790TIA

Kaushlendra Singh Sisodia
Chief Mentor, The IoT Academy
Director, Uniconverge Technologies

Internship Offer Letter

Date: **25-07-2026**

Name: **Sachin Kumar**

Email: **sachin.25scs1003004017@iilm.edu**

Location: **Remote**

Dear **Sachin Kumar**,

We are pleased to inform you that **Upskill Campus** alongwith its industry Partner **UniConverge Technologies Pvt. Ltd.** has agreed to provide you internship as **Data Science and Machine Learning Intern**

Your appointment of service is based on the following terms and conditions.

1. The internship shall start from **25-07-2026**
2. **Internship:** The internship period will be 4 to 6 weeks, starting from the date of joining.
3. **Working Hours:** The internship runs from 10:00 AM to 6:00 PM.

On behalf of both the Company, we wish you every success in your position and trust that our relationship will be long and mutually rewarding.

Sincerely

For Upskill Campus and UniConverge Technologies Pvt. Ltd.



Prabha Singh
HR Manager